Cyclone Remal (2024): effect of prescribed OSCAR surface currents on WAVEWATCH III waves

Main-paper results compilation. Analysis window 2024-05-24 00:00 to 2024-05-28 00:00 UTC (inclusive). Common sample: 15,635 altimeter 1-Hz observations from 67 passes.

The analysis window ends at 28 May 00 UTC to keep the evaluation focused on the Remal event and avoid mixing the targeted storm-period comparison with the later post-storm wave field.

Differences between WW3 OSCAR and WW3 CONTROL are interpreted as the net response to prescribed surface-current forcing in this configuration (RWND active), not a pure refraction effect. ERA5 is a reanalysis benchmark, not independent, because it shares ERA5 winds and boundary spectra with the WW3 runs. Figure 4 is Hisaki-inspired, not a replication.

Table 1. Model configuration and datasets

Group	Item	Value
Model	Wave model	WAVEWATCH III 6.07.1 (RWND build option ACTIVE)
Model	Domain / grid	75-100 E, 0-25 N; regular spherical 0.10 deg; 251 x 251 points
Model	Bathymetry	GEBCO 2026
Model	Wind forcing	ERA5 hourly 10-m winds
Model	Open boundaries	ERA5 two-dimensional directional wave spectra
Model	Spectral grid	32 frequencies from 0.04118 Hz (ratio 1.10); 36 directions; offset 0 deg
Model	Physics	ST4 source terms; NL1, BT4, DB1 switches; RWND (current-relative wind) active
Model	Time steps	global 600 s; spatial 300 s; spectral 300 s; source 150 s
Model	Water-level forcing	Off
Model	Initial condition	Same calm restart file for both experiments
Model	Production run	20 May 00 UTC - 29 May 23 UTC 2024 (first 96 h spin-up)
Experiments	CONTROL	No prescribed surface currents
Experiments	OSCAR	Identical configuration + prescribed OSCAR L4 Final V2 total surface currents (0.25 deg, daily)
Reference	Observations	1-Hz multi-mission satellite altimeter Hs (independent reference)
Reference	ERA5 wave benchmark	ERA5 significant height of combined wind waves and swell; reanalysis benchmark, not independent (it shares ERA5 winds and boundary spectra with the WW3 runs)
Analysis	Frozen window	24 May 2024 00 UTC - 28 May 2024 00 UTC (inclusive)
Analysis	Alignment variable	U_parallel = u*sin(theta_to) + v*cos(theta_to), theta_to = (CONTROL mean wave dir FROM + 180) mod 360; <0 opposing, >0 following
Data in this paper	Altimeter observations in window (missions used)	16,347
Data in this paper	Common comparison sample	15,635 1-Hz observations from 67 passes

Data in this paper	Missions in common sample	CryoSat-2, HY-2B, HY-2C, Jason-3, SARAL/AltiKa, SWOT-nadir, Sentinel-3A, Sentinel-3B, Sentinel-6A
Data in this paper	ERA5 / CONTROL / OSCAR	all three evaluated on exactly the same observations
Data in this paper	IMD track	28 original records + 69 linearly interpolated hourly positions (24-28 May); interpolated rows are not observations

Figure 1. Study area

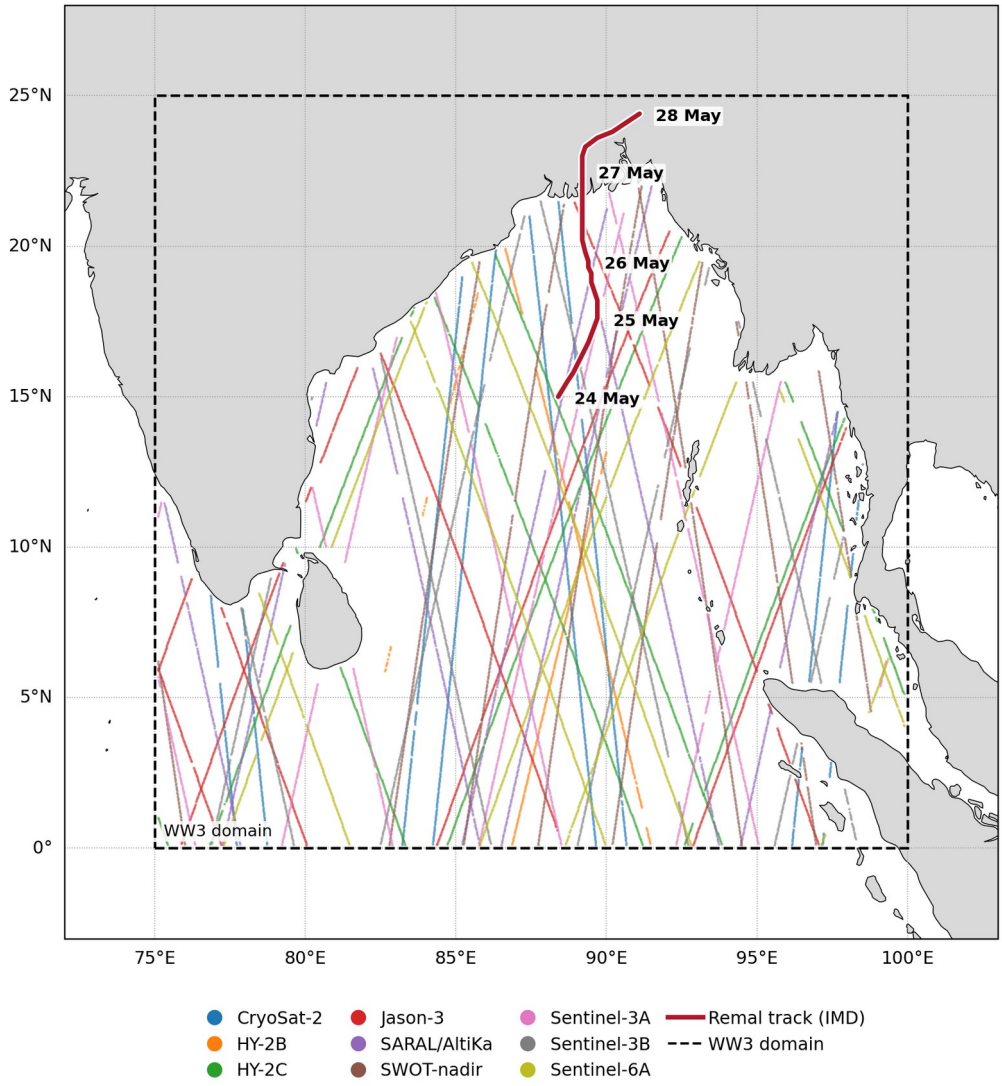


Figure 1. Study area. Dashed box: WAVEWATCH III domain. Solid line: hourly Cyclone Remal track (IMD), 24 May 00 UTC to 28 May 00 UTC 2024. Small dots show altimeter 1-Hz sampling of the common comparison sample (N = 15,635), coloured by mission. Grey: land (Natural Earth coastlines).

Figure 2. Hs validation

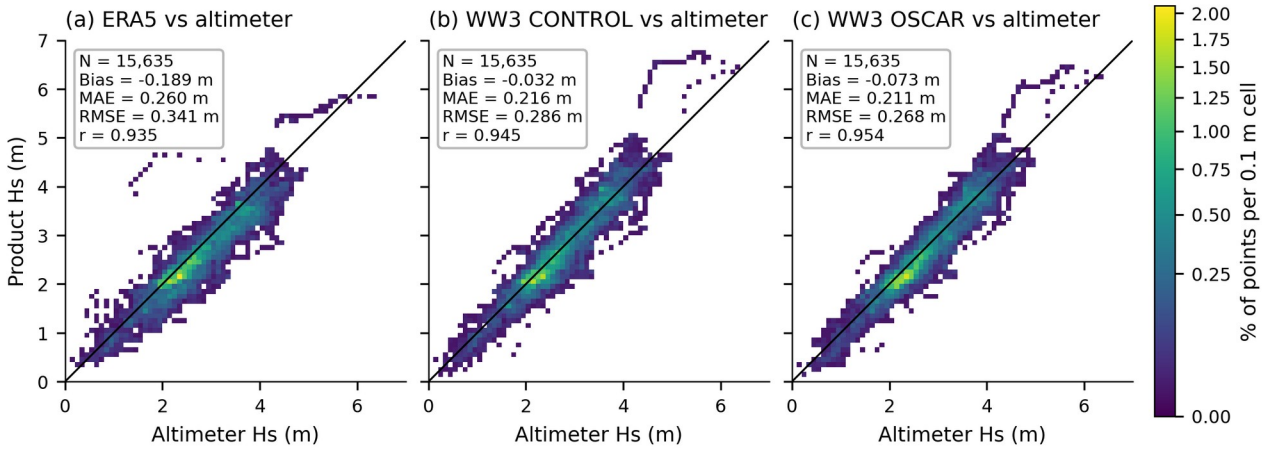


Figure 2. Significant wave height against satellite altimeters on the common sample: (a) ERA5 reanalysis benchmark, (b) WW3 CONTROL, and (c) WW3 OSCAR. Colours show the percentage of points per 0.1 m cell; the line is 1:1. Bias = model - observation.

Table 2. Common-sample validation

Product	N	Bias (m)	MAE (m)	RMSE (m)	r
ERA5	15,635	-0.189	0.260	0.341	0.935
WW3 CONTROL	15,635	-0.032	0.216	0.286	0.945
WW3 OSCAR	15,635	-0.073	0.211	0.268	0.954

Figure 3. Direct WW3 response

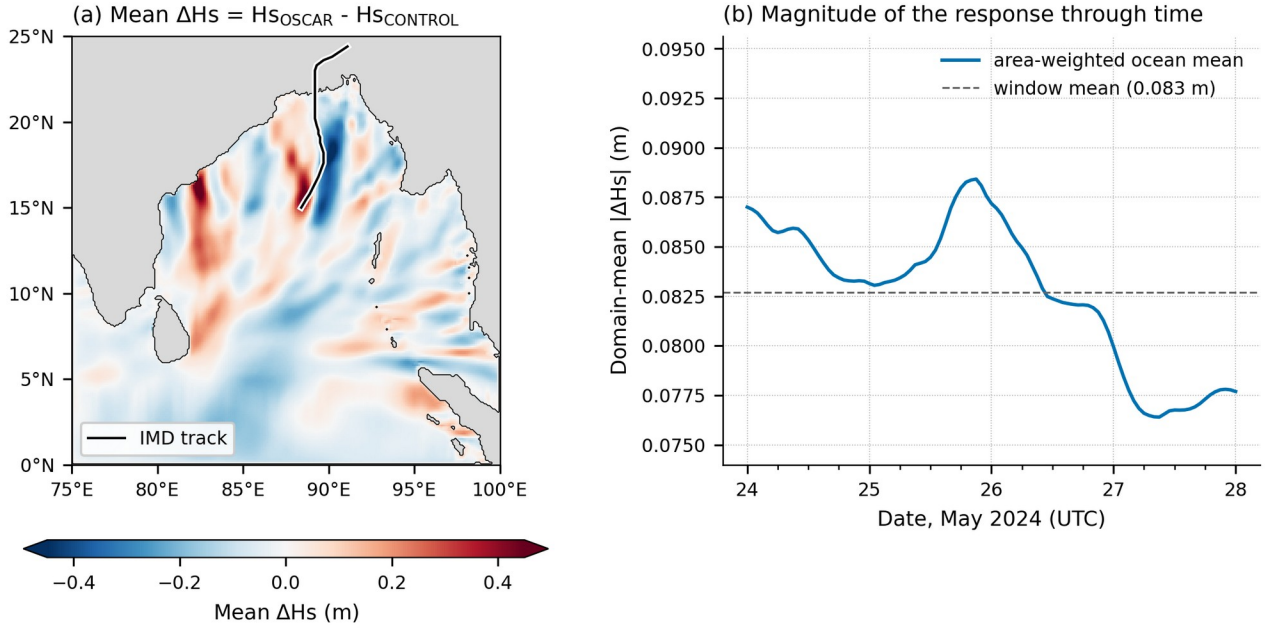


Figure 3. Net response of the WW3 wave field to prescribed OSCAR surface-current forcing, 24 May 00 UTC to 28 May 00 UTC 2024. (a) Time-mean Delta Hs = Hs(OSCAR) - Hs(CONTROL) with the

IMD track. (b) Area-weighted domain-mean absolute Delta Hs per hour. RWND was active, so the difference includes any current-relative-wind effect.

Figure 4. Current-direction dependence

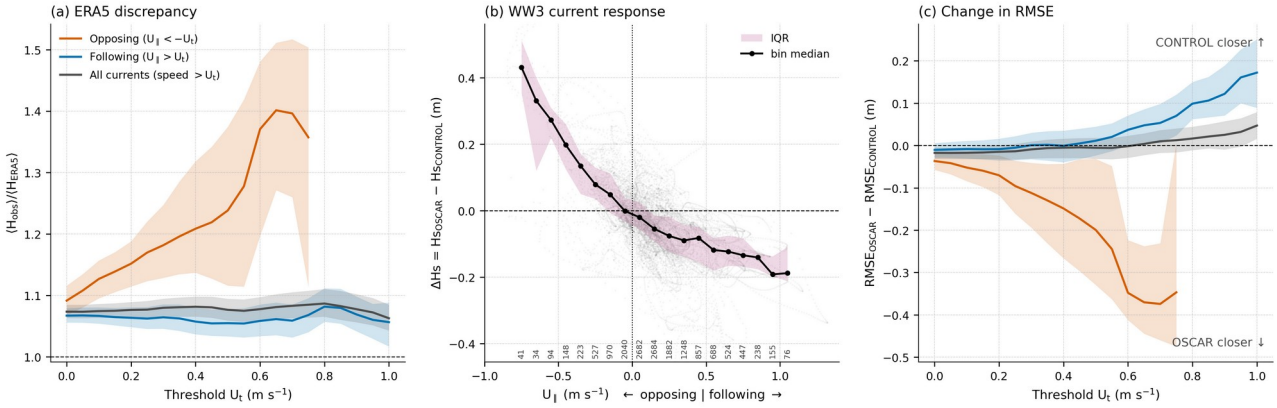


Figure 4. Hisaki-inspired analysis of current dependence. U_parallel is the OSCAR current component along the WW3 CONTROL mean wave-propagation direction (negative: opposing; positive: following). (a) Ratio of mean altimeter Hs to mean ERA5 Hs for cumulative thresholds. (b) Delta Hs = Hs(OSCAR) - Hs(CONTROL) in 0.1 m/s U_parallel bins; line = median and band = IQR. (c) RMSE(OSCAR) - RMSE(CONTROL), where values below zero mean OSCAR is closer to the altimeter. Confidence bands in (a) and (c) resample altimeter passes.

Table 4. Pass-level robustness

Statistic	Value
Passes in common sample	67
Valid passes (at least 10 points)	66
Mean pass MAE, CONTROL (m)	0.206
Mean pass MAE, OSCAR (m)	0.208
Mean Delta MAE_pass (m) [95% bootstrap CI]	0.0013 [-0.0073, 0.0099]
Median Delta MAE_pass (m) [95% bootstrap CI]	0.0054 [-0.0093, 0.0158]
Passes where OSCAR has the smaller MAE	30 of 66 (45.5%)
Two-sided sign test p-value	0.539
Bootstrap	5,000 resamples of passes (seed 43)

Delta MAE_pass = MAE_OSCAR - MAE_CONTROL. Negative values mean OSCAR is closer to the altimeter. Passes, rather than individual 1-Hz points, are the bootstrap units.